Topic 1: Recap from management accounting

Sessions 1-2

Carl Lewis?



Power without control is useless (Pirelli 1994)



AN ICON THAT IS STILL RELEVANT

Of the many images that have depicted Pirelli over the years, perhaps one of the most extraordinary is that of American sprinter Carl Lewis in the typical starting position wearing a pair of red stiletto heels. Already a world record holder and having won his eighth Olympic gold medal, Carl Lewis and those heels perfectly embodied the message that Pirelli wanted to convey: power is nothing without control. Twenty-five years on, those words are more relevant than ever before.

The 2018 Annual Report celebrates this with a short film, a photo essay and the reflections of three authors who each offer their personal insight into the famous payoff, presented from their own unique point of view.

Learning objectives



- If you look at picture A, you will realize that the dog guarding the sheep has targeted a sheep.

Learning objectives



- If you look at picture B, you will realize that the dog guarding the sheep has pulled a sheep out of the water, saving it from drowning.

- Just remember that the better the knowledge and information we get, the closer we will be to reality, and no one can deceive a realistic person. And the best knowledge is not based on hearsay.

Learning objectives

- Management accounting as a key player in the business
- Regardless you are or not an accountant, you MUST understand the figures
- Understanding the cost structure to drive the strategy
- Simple and ABC cost accounting systems
- Activity drivers
- Cost hierarchy
- Pros and cons of adopting ABC

“Accounting”

- Financial accounting focuses on reporting financial information to external parties such as investors, government agencies, banks, and suppliers based on Generally Accepted Accounting Principles (GAAP).

- Management accounting is the process of measuring, analyzing, and reporting financial and nonfinancial information that helps managers make decisions to fulfill the goals of an organization.

- Cost accounting provides information for both management accounting and financial accounting professionals. Cost accounting is the process of measuring, analyzing, and reporting financial and nonfinancial information related to the costs of acquiring or using resources in an organization.

“Accounting”

EXHIBIT 1-1 Major Differences Between Management and Financial Accounting

	Management Accounting	Financial Accounting
Purpose of information	Help managers make decisions to fulfill an organization’s goals	Communicate an organization’s financial position to investors, banks, regulators, and other outside parties
Primary users	Managers of the organization	External users such as investors, banks, regulators, and suppliers
Focus and emphasis	Future-oriented (budget for 2017 prepared in 2016)	Past-oriented (reports on 2016 performance prepared in 2017)
Rules of measurement and reporting	Internal measures and reports do not have to follow GAAP but are based on cost-benefit analyses	Financial statements must be prepared in accordance with GAAP and be certified by external, independent auditors
Time span and type of reports	Varies from hourly information to 15 to 20 years, with financial and nonfinancial reports on products, departments, territories, and strategies	Annual and quarterly financial reports, primarily on the company as a whole
Behavioral implications	Designed to influence the behavior of managers and other employees	Primarily reports economic events but also influences behavior because manager’s compensation is often based on reported financial results

Traditional (old) cost accounting

- Historically, companies produced a limited variety of products.
- These companies used few overhead resources to support these simple operations, so indirect (or overhead) costs were a relatively small percentage of total costs.
- Managers used simple costing systems to allocate overhead costs broadly in an easy, inexpensive, and reasonably accurate way.
- But as product diversity and indirect costs increased, broad averaging led to inaccurate product costs. That's because simple peanut-butter costing (that's what it's called) broadly averages or spreads the cost of resources uniformly to cost objects (such as products or services) when, in fact, the individual products or services use those resources in nonuniform ways.

Traditional (old) cost accounting

- Under costing and over costing. Broad averaging often leads to under costing or over costing of products or services:
  - Product under costing: a product is reported to have a low cost per unit but consumes a higher level of resources per unit.
  - Product over costing—a product is reported to have a high cost per unit but consumes a lower level of resources per unit.

- Under cost products will be underpriced and may even lead to sales that actually result in losses because the sales may bring in less revenue than the cost of resources they use.

- Over cost products will lead to overpricing, causing those products to lose market share to competitors producing similar products.

Traditional (old) cost accounting

- The following example illustrates how averaging can result in inaccurate and misleading cost data (Undercost: James. Overcost: Emma)

	Emma	James	Jessica	Matthew	Total	Average
Entree	$11	$20	$15	$14	$ 60	$15
Dessert	0	8	4	4	16	4
Drinks	4	14	8	6	32	8
Total	$15	$42	$27	$24	$108	$27

- Product-cost cross-subsidization means that if a company undercosts one of its products, it will overcost at least one of its other products. Similarly, if a company overcosts one of its products, it will undercost at least one of its other products. Product-cost cross-subsidization is very common when a cost is uniformly spread—meaning it is broadly averaged—across multiple products without managers recognizing the amount of resources each product consumes.

Refining the costing system

There are three main guidelines for refining a costing system:

1. **Direct-cost tracing**. Identify as many direct costs as is economically feasible.

2. **Indirect-cost pools**. Expand the number of indirect-cost pools until each pool is more homogeneous. All costs in a homogeneous cost pool have the same or a similar cause-and-effect (or benefits-received) relationship with a single cost driver that is used as the cost-allocation base.

3. **Cost-allocation bases**. Whenever possible, managers should use the cost driver (the cause of indirect costs) as the cost-allocation base for each homogeneous indirect-cost pool (the effect).

ABC: Activity Based Costing

One of the best tools for refining a costing system is activity-based costing. Activity-based costing **(ABC) refines a costing system by identifying individual activities as the fundamental cost objects.** An activity is an event, task, or unit of work with a specified purpose—for example, designing products, setting up machines, operating machines, or distributing products.

To help make strategic decisions, ABC systems **identify activities in all functions of the value chain, calculate costs of individual activities, and assign costs to cost objects** such as products and services on the basis of the mix of activities needed to produce each product or service.

The logic of ABC systems is twofold:

1) First, when managers structure activity cost pools more finely, using cost drivers for each activity cost pool as the **cost-allocation base**, it leads to more accurate costing of activities.
2) Allocating these costs to products by measuring the cost allocation bases of different activities used by different products **leads to more accurate product costs**.

Cost hierarchies (I)

A cost hierarchy categorizes various activity cost pools on the basis of the different types of cost drivers, cost-allocation bases, or different degrees of difficulty in determining cause-and-effect (or benefits-received) relationships.

ABC systems commonly use a cost hierarchy with **four levels to identify cost-allocation bases that are cost drivers of the activity cost pools**:

1) **Output unit–level costs**. The costs of activities performed on each individual unit of a product or service. Machine operations costs (such as the cost of energy, machine depreciation, and repair) related to the activity of running the automated molding machines are output unit–level costs because, over time, the cost of this activity increases with additional units of output produced (or machine-hours used).

Cost hierarchies (II)

(2) Batch-level costs. The costs of activities related to a group of units of a product or service rather than each individual unit of product or service. Example: the cost of the machine setup activity varies with the setup hours needed to produce batches (groups) of different products regardless of the total number of products produced.

(3) Product-sustaining costs. The costs of activities undertaken to support individual products or services regardless of the number of units or batches in which the units are produced or services provided. Example: cost of designing a product.

(**4) Facility-sustaining costs**. The costs of activities that managers cannot trace to individual products or services but that support the organization as a whole. Example: headquarters admin costs.

Advantages of abc cost allocation

- Difference between “traditional” and ABC costing (Activity, rather than production volume, causes costs)
- To know the REAL cost of a product/Service
- To know gross margins of products/services
- Identify profitable products/services
- Compare to competitor’s gross margins
- Allocate OH accurately to the different products

Overhead rate

$$\text{OH RATE} = \frac{\text{TOTAL OH COST}}{\text{TOTAL VOLUME OF COST DRIVER}}$$

Example:

- Total OH Cost = 250,000 €
- Total machine hours to be used = 10,000 machine hours
- OH rate = 25 € / machine hour

Practical case: Plastim Corp.

- Plastim Corporation manufactures lenses for the rear taillights of automobiles.
- Plastim manufactures two types of lenses for Giovanni Cars: a simple lens called S3 and a complex lens called C5.
- Giovanni's purchasing manager informed Plastim's sales manager that a competitor (Bandix), is offering to supply the S3 lens to Giovanni at a price of $53, well below the $63 price that Plastim is currently projecting. Unless Plastim can lower its selling price, it will lose the Giovanni business for the simple lens for the upcoming model year.
- Fortunately, the same competitive pressures do not exist for the complex lens, which Plastim currently sells to Giovanni at $137 per lens.
- Plastim's managers have two primary options:
  - Give up the Giovanni business in simple lenses if selling them is unprofitable.
  - Reduce the price of the simple lens and either accept a lower margin or aggressively seek to reduce costs.

Practical case: Plastim Corp.

A) Calculate full unit cost using a traditional costing system where overhead cost driver are direct labor hours.
B) Calculate full unit cost and gross margin using ABC costing system.

	S3	CL5
Selling price ($)	63,00	137,00
Units produced	60.000,00	15.000,00
# of labor hours required per unit	0,50	0,65
Price ($) per hour of direct labor	20,00	20,00
Direct material costs ($) per unit	18,75	45,00

OVERHEAD COSTS	Total cost	Usage by S3	Usage by	TOTAL	Comments (cost driver)
Mold and maintenance labor	270.000,00	120.000,00	150.000,00		Total number of units
Design	450.000,00	0,30	0,70	1,00	Total parts per square feet
Setup molding machines	300.000,00	500 SH	1,500 SH	2.000,00	Total set up hours
Machine operations	637.500,00	9,000 MH	3,750 MH	12.750,00	Total machine operations
Shipment setup	81.000,00	100 S	100 S	200	Number of shipments
Distribution	391.500,00	45,000 ft^3	22,500 ft^3	67.500,00	Total cubic feet delivered
Administration costs	255.000,00	30,000 LH	9,750 LH	39,750 LH	Total direct labor hours
TOTAL OVERHEAD	2.385.000,00				

Practical case: Plastim Corp.

A) Calculate full unit cost using a traditional costing system where overhead cost driver are direct labor hours.

A: Driver OH = Labour costs	S3	CL5	Total
Sales	3.780.000,00	2.055.000,00	5.835.000,00
Direct Cost	1.725.000,00	870.000,00	2.595.000,00
# hours	30.000,00	9.750,00	39.750,00
OH rate	75%	25%	
OH allocation	1.800.000,00	585.000,00	2.385.000,00
Total costs	3.525.000,00	1.455.000,00	4.980.000,00
Cost per unit	58,75	97,00	
Gross margin	255.000,00	600.000,00	855.000,00
Margin per unit	4,25	40,00	11,40

	S3	CL5	Total
# of labor hours required per unit	0,50	0,65	
Price ($) per hour of direct labor	20,00	20,00	
Direct material costs ($) per unit	18,75	45,00	
Units produced 2011	60.000,00	15.000,00	75.000,00
# of labor hours	30.000,00	9.750,00	39.750,00
OH allocation rate	0,75	0,25	
OH per unit	30,00	39,00	2.385.000,00
OH per product (total)	*1.800.000,00*	*585.000,00*	*2.385.000,00*
Direct labour cost / unit	10,00	13,00	
Direct material cost / unit	18,75	45,00	
OH per unit	30,00	39,00	
Cost per unit	**58,75**	**97,00**	
Selling price ($)	63,00	137,00	
Margin per unit	4,25	40,00	

Practical case: Plastim Corp.

B) Calculate full unit cost and gross margin using ABC costing system.

				(Per unit)	
B: (ABC)	**S3**	**CL5**	**Total**	**S3**	**CL5**
Sales	3.780.000,00	2.055.000,00	5.835.000,00	63,00	137,00
# hours	30.000,00	9.750,00	39.750,00	0,50	0,65
Direct labour cost	600.000,00	195.000,00	795.000,00	10,00	13,00
Direct material cost	1.125.000,00	675.000,00	1.800.000,00	18,75	45,00
Diect cost	**1.725.000,00**	**870.000,00**	**2.595.000,00**	**28,75**	**58,00**
OH					
Mold and maintenance labor	120.000,00	150.000,00	270.000,00	2,00	10,00
Design	135.000,00	315.000,00	450.000,00	2,25	21,00
Setup molding machines	75.000,00	225.000,00	300.000,00	1,25	15,00
Machine operations	450.000,00	187.500,00	637.500,00	7,50	12,50
Shipment setup	40.500,00	40.500,00	81.000,00	0,68	2,70
Distribution	261.000,00	130.500,00	391.500,00	4,35	8,70
Administration costs	192.452,83	62.547,17	255.000,00	3,21	4,17
OH allocation	**1.273.952,83**	**1.111.047,17**	**2.385.000,00**		
OH per unit	21,23	74,07	31,80	**21,23**	**74,07**
Total costs	2.998.952,83	1.981.047,17	4.980.000,00		
Cost per unit	49,98	132,07	66,40	49,98	132,07
Gross margin	781.047,17	73.952,83	855.000,00		
Margin per unit	**13,02**	**4,93**	**11,40**	**13,02**	**4,93**

Takeaways

Broad averaging (or "peanut-butter costing") describes a costing approach that uses broad averages for assigning (or spreading, as in spreading peanut butter) the cost of resources uniformly to cost objects.

Over costing may result in overpricing and competitors entering a market and taking market share for products that a company erroneously believes are low-margin or even unprofitable. Under costing may result in companies selling products on which they are in fact losing money, when they erroneously believe them to be profitable.

Costing system refinement means making changes to a simple costing system that reduces the use of broad averages for assigning the cost of resources to cost objects and provides better measurement of the costs of overhead resources used by different cost objects. Three guidelines for refinement are:

1. Classify as many of the total costs as direct costs as is economically feasible.
2. Expand the number of indirect cost pools until each of these pools is more homogenous.
3. Use the cause-and-effect criterion, when possible, to identify the cost-allocation base for each indirect-cost pool.

Takeaways

An ABC approach focuses on activities as the fundamental cost objects. The costs of these activities are built up to compute the costs of products, and services, and so on.

An ABC approach attempts to use cost drivers as the allocation base for indirect costs, whereas a simple costing system generally does not. The ABC approach classifies as many indirect costs as direct costs as possible.

Cost allocation starts with the distinction between direct and indirect costs. When both direct and indirect costs are calculated for a cost object, then the full cost of a cost object is calculated. The full cost principle indicates the use of absorption costing, as shown in the following matrix:

	Variable costs	Fixed costs	Total
Direct costs			Total direct costs
Indirect costs			Total indirect costs
Total	Total variable costs	Total fixed costs	Total costs

Takeaways

The four levels of indirect costs within ABC are: output unit level costs, batch level costs, product sustaining costs, and facility sustaining costs. Therefore, the cost objects for these four levels are:

ABC level costs	Cost object
Unit level	Unit of product or service
Batch level	Group of product or services
Product level	Group of products or batches
Facility level	The organization

Practical case: Quickprint (ABC costing)

6. **Quickprint. Inc.** prints leaflets, brochures and advertising material. The company classifies its various printing jobs as standard jobs and special jobs. Quickprint job-costing has 2 direct cost categories (direct material and direct labor). Quickprint is planning to implement an Activity Based Costing system. The following table shows information for 2015 fiscal year:

	Standard	Special
# Printing jobs	400,00	200,00
Price per job	1.200,00	1.500,00
Cost of suppliers per job	200,00	250,00
Direct labor cost/job	180,00	200,00
Printing machine hour per job	10,00	10,00

Practical case: Quickprint (ABC costing)

OVERHEAD INFO	Standard	Special	Total cost	Cost driver
Set-up hours / job	4,00	7,00		
Set-up costs			90.000,00	Set-up hours
Cost printing machine			150.000,00	Machime hours
Total # of P.O.	400,00	500,00		
P.O. costs			36.000,00	# P.O.
Desing costs	8.000,00	32.000,00		
Admin costs			90.000,00	Direct labor costs

Calculate the cost per unit and gross margin for product of both types of jobs using the ABC system

Practical case: Quickprint (ABC costing)

Costs breakdown	Standard	Special	
Cost of suppliers per job	**200,00**	**250,00**	
Direct labor cost/job	**180,00**	**200,00**	
Set-up hours / job	4,00	7,00	
# set-up hours	1.600,00	1.400,00	3.000,00
Set-up costs	48.000,00	42.000,00	**90.000,00**
Set-up costs per job	**120,00**	**210,00**	
Printing machine hours per job	10,00	10,00	
# printing hours	4.000,00	2.000,00	6.000,00
Cost printing machine	100.000,00	50.000,00	**150.000,00**
Cost printing per job	**250,00**	**250,00**	
Total # of P.O.	400,00	500,00	900,00
P.O. costs	16.000,00	20.000,00	**36.000,00**
P.O. costs per job	**40,00**	**100,00**	
Desing costs	8.000,00	32.000,00	
Desing costs per job	**20,00**	**160,00**	
Direct labor cost/job	180,00	200,00	
Direct labor cost (total)	72.000,00	40.000,00	112.000,00
Admin costs	57.857,14	32.142,86	**90.000,00**
Admin cost per job	**144,64**	**160,71**	
Total cost/job	**954,64**	**1.330,71**	
Sales price /job	1.200,00	1.500,00	
Gross margin	**245,36**	**169,29**	

Total set-up hours	1.600,00	1.400,00	3.000,00
Cost / hour			30,00
# Printing jobs	400,00	200,00	
Hour per job	4,00	7,00	
Cost per job	120,00	210,00	

Practical case: Quickprint (ABC costing)

	Standard job	**Special job**
Direct materials	200 €/job	250 €/job
Labor costs	180 €/job	200 €/job
Set-up costs	30€/h*4 h/job = 120 €/job	30 €/h*7h/job=210 €/job
Printing Machine costs	25€/MH*10MH/job= 250€/job	25€/MH*10MH/job= 250€/job
Purchase order costs	40€/po*400po/400job= 40€/job	40€/po*500po/200job= 100€/job
Design costs	8000€/400 jobs= 20 €/job	32000€/200 jobs= 160 €/job
Administrative costs	0.8036 €/€*180 €/job= 144.65 €/job	0.8036 €/€*200 €/job= 160.72 €/job
TOTAL COST PER JOB	**954.65 €/job**	**1330.72 €/job**
PRICE PER JOB	1200 €/job	1500 €/job
GROSS MARGIN	**245.35 €/job**	**169.28 €/job**

Practical case: PC and Cells Co. (ABC costing)

A company produces two products: a computer part and cell phones. Overhead costs are currently allocated based on direct labor hours. Calculate the cost of each product allocating overhead using direct labor hours as cost driver and then using an ABC system.

Direct Costs	**PCs**	**Phones**
Units to be produced in 2015	4.000,00	2.000,00
Direct labor hours per unit	2,00	0,50
Direct material cost ($) per unit	30,00	17,00
Direct labor cost ($) per unit	16,00	4,00

TOTAL FACTORY OVERHEAD

Activity	**OH Costs**	**Tot. volume**	**PCs**	**Phones**	
Production setups	80.000,00	20,00	5,00	15,00	Setups
Material handling	70.000,00	5.000,00	1.000,00	4.000,00	lbs
Packaging and shipping	120.000,00	6.000,00	4.000,00	2.000,00	boxes
Total OH	**270.000,00**				

Practical case: PC and Cells Co. (ABC costing)

Cost per unit using direct labor hours for OH allocation

	PCs	Phones	
Direct labor hours per unit	2,00	0,50	
Total labor hour used	8.000,00	1.000,00	9.000,00
OH rate	88,9%	11,1%	
Total OH	240.000,00	30.000,00	
OH per unit	60,00	15,00	
Direct material cost ($) per unit	30,00	17,00	
Direct labor cost ($) per unit	16,00	4,00	
OH	60,00	15,00	
Total cost per unit	**106,00**	**36,00**	

Practical case: PC and Cells Co. (ABC costing)

Cost per unit using ABC

Activity	**OH Costs**	**Volume**	**PCs**	**Phones**	
Production setups	80.000,00	20,00	5,00	15,00	Setups
Material handling	70.000,00	5.000,00	1.000,00	4.000,00	lbs
Packaging and shipping	120.000,00	6.000,00	4.000,00	2.000,00	boxes
Total OH	**270.000,00**				
Units		*6.000,00*	*4.000,00*	*2.000,00*	
Production setups	80.000,00		5,00	30,00	
Material handling	70.000,00		3,50	28,00	
Packaging and shipping	120.000,00		20,00	20,00	
OH allocation per unit			**28,50**	**78,00**	
Direct cost per unit			46,00	21,00	
Total cost per unit			**74,50**	**99,00**	

Practical case: Environmental

A company produces two types of microprocessors, A & B. Generally, the company has considered all environmental costs as overhead.
Lately, they have decided to use ABC costing to allocate 3 environmental costs to the two products, as each of them generates different quantities of environmentally related costs.
The three costs relate to air pollution, to waste recycling, and to monitoring environmental effects.

Calculate the cost to produce one unit of each microprocessor

	Microprocessor A	**Microprocessor B**
Number of microprocessors produced	2.000,00	1.500,00
Direct material cost per microprocessor	2,10	2,30
Direct labor cost per microprocessor	1,50	1,50
Direct labor hours per microprocessor	5,90	6,20
Tons of recycled general waste per microprocessor	**0,09**	**0,097**
Number of internal audits	1,00	2,00

Environmental Overhead Costs		
	Total cost	**Cost driver**
Cost of absenteeism (due to air pollution)	30.000,00	Direct labor hours
Cost of recycling general waste	9.350,00	Tons of recycled waste
Costs of environmental auditing	5.480,00	# of internal audits

Practical case: Environmental

	Microprocessor A	Microprocessor B			A	B	
Number of microprocessors produced	2.000,00	1.500,00	3.500,00		57,1%	42,9%	
Direct material cost per microprocessor	2,10	2,30			**2,10**	**2,30**	
Direct labor cost per microprocessor	1,50	1,50			**1,50**	**1,50**	
Direct labor hours per microprocessor	5,90	6,20					
Tons of recycled general waste per microprocessor	**0,09**	**0,097**					
Number of internal audits	1,00	2,00					
					3,60	**3,80**	
Environmental Overhead Costs							
	Total cost	**Cost driver**					
Cost of absenteeism (due to air pollution)	30.000,00	Direct labor hours	8,57		8,39	8,82	
Cost of recycling general waste	9.350,00	Tons of recycled waste	2,67		2,59	2,79	
Costs of environmental auditing	5.480,00	# of internal audits	1.826,67		0,91	2,44	
Labour hours	11.800,00	9.300,00	21.100,00		**11,89**	**14,04**	
Cost of absentism	16.777,25	13.222,75	30.000,00				
Cost of absentism per unit	**8,389**	**8,815**	8,57		**15,49**	**17,84**	**Cost/unit**
Tons of waste	180,00	145,50	325,50				
Cost of recycling	5.170,51	4.179,49					
Cost of recycling per unit	**2,59**	**2,79**					

Takeaways

Takeaways

- A cost system is mostly about **allocating overhead costs** (indirect costs) to products
- ABC costing is about estimating the overhead transactions (activities) caused by each product (cost object)
- Different cost allocation systems cause different profitability estimations
- A cost that can be assigned to a product line (direct cost) should be assigned that way
- Many companies continue to use direct labor to allocate overhead
- Companies may not allocate **all** costs to products

References

- *Horngren et al.: Chapter 5*